Impact Evaluator Design - Juan Benet

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CryptoEconLab has:

- Designed a first functional two-sided decentralized marketplace and island economy with its own full-fledged economic policy
- Evaluated and implemented the first deployment of EIP1559 with multiple modifications and innovations
- Championed scalable Layer 1 research and its unique and lesser known economic challenges
- Pioneered data-driven analytics for community governance and network evolution
- Positioned to become the data availability layer for other Web3 blockchains with its scale and verifiability
- Introduced unique building blocks to Web3 with immense new business opportunities: reputation, retrieval markets, DataDAOs, Compute-over-Data, etc.

This FIL Austin satellite event will provide a deep dive into Filecoin’s future developments. Join us for a fun and educational look into the future of the fast-paced world of cryptoeconomics.

Transcript

0:00

a lot of thoughts on crypto econ um one thing i wanted to quickly mention um is that there's a great

0:07

uh book and um uh called green build that kevin iwaki

0:12

and the and the folks in the githune community have been putting together that is has a very good introduction to a lot of

0:20

the concepts about how crypto econ can create regenerative systems

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and build structures that are positive some in the broader world and can tackle large-scale problems so

0:32

highly recommend like looking at that whole meme complex and um exploring

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the world of crypticon through the lens of regenerative growth economics regenerative finance

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and so on um and the kind of like the core message of all of that is that at the end of the day all

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of the all of our macro systems and our governance systems and so on are about how to

0:57

coordinate large groups of participants towards some set of shared goals

1:03

and of course there's like tons of layers of complexity there when you think about different preferences and different

1:08

goals and so on different values but it's all kind of coordination systems at the end of the day

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so if you kind of step back and look at macro systems that way you can then start kind of decomposing

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them and figuring out what structures can yield better

1:25

coordination principles better coordination structures to achieve better outcomes together

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and the kind of like amazing prominence of our time today and this is kind of why

1:36

i like thinking about it in terms of solving planetary scale problems

1:43

is that you can use these kinds of structures these kind of cryptoecon structures to

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have extremely large scale impact and just to give a sense of how large scale

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that impact can be just think of something like the bitcoin network or the falcoin storage providers

2:04

in a very short span of time these networks have assembled massive scale

2:10

computing power and massive scale service provider

2:16

it's a massive skill network with an enormous amount of resource consumption and uh

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and utility provided and organized by a few mechanisms so what i wanted to kind of

2:29

um i want to spend most of the time today talking about a core component

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in these which um we're sort of describing as impact

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evaluators um i'm not familiar with this with this kind of mechanism description um being applied before um so we sort of

2:47

like coined it but i kind of want to dive into together as a group into

2:52

potential designs for impact evaluators uh and then if they're if they're sort of

2:58

like um uh time at the end i want to kind of then uh talk about kind of how you can

3:05

use crypto econ systems to kind of solve larger scale problems um i gave a talk like this at the last crypto econ

3:11

day that kind of goes into falcon green and looks at that as a project and it's a system that sort of

3:18

tries to tackle a large scale problem breaks it down into smaller and smaller smaller components

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and then tries to create incentive structures to solve each one of those problems

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to then sort of vault over time into a larger more coordinated group um

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solving each kind of progressively larger problem so as an example if you want to

3:41

uh decarbonize the planet you can start by the carbonate decarbony decarbonizing an industry

3:47

um and if you want to decarbonize an industry you could pick something like cloud computing you could then um create

3:54

a substance and say great let's decarbonize first crypto cloud computing and start with

4:00

a storage network and then from there start with the file coin storage network and you can each time as you kind of

4:06

decompose a problem start with a smaller set that is much more achievable figure out the system structures create

4:13

positive some incentive structures to achieve that outcome and then use that as an example for them the layer above

4:19

right so if we can get to fully decarbonize falcoin then we can use that as an example to

4:25

other crypto networks to decarbonize themselves once we achieve that and create a strong

4:30

incentive structure to get all these participants doing that we can then vault that into

4:36

talking to other industries and saying hey look the crypto industry has now fully carbonized it is net green

4:44

this is totally achievable hey other industries you go do it too uh once you get many more industries doing this then

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you can slowly get you know chip away at the problem so um the

4:57

i've come to think that that uh crypto econ is kind of like the highest leverage um set of tool sets that

5:03

humanity has at the moment to solve some of the largest problems we have because they it lets us break down systems

5:11

design mechanisms and deploy them at scale quickly um

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and if they're working well they can scale quickly into some pretty enormous

5:23

skills so the uh let's see

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the kind of picture that i want people to kind of consider is this for any problem that you might be

5:35

running into trying to understand what the incentive landscape looks like try to

5:41

think about what uh what structures and mechanisms are causing what kind of actions by what parties

5:47

if you are trying to get to a different outcome think of the barriers in between like what are the hills that are preventing

5:53

um that are locking us into a an inadequate equilibrium here and think of

5:58

creating mechanisms and structures to warp the incentive field to get to that better equilibrium

6:03

and you can do this progressively so you can you know start with you know one such problem space you can think of like

6:10

using um mechanism design as like some tool that would let you either like tunnel through a mountain in that

6:16

incentive landscape or um ideally you want to like pinch the landscape and like move it down but that's kind of like

6:22

a harder visual metaphor and then over time as you kind of like solve more and more problems you can

6:27

progressively come up with other structures and so on and steer the

6:33

planet into a much better much better condition i think along the way

6:41

what's what's different now about the the world today post blockchains from

6:48

the world before is that what's really going on is that software is eating mechanism design so

6:56

think of the computing platform that we've deployed as building this

7:01

extremely programmable environment where you can deploy any kind of software and deploy it to supercomputers running

7:06

everywhere including our pockets and our wrists and whatever soon to be

7:12

all over you we have trillions of devices and so on and we have an extremely upgradable

7:18

environment where individuals and small groups of people can dream up some new structure and

7:25

deploy it out into the world and get a very high quality

7:30

production feedback loop of whether or not that thing is good how well it works and so on to the point where you can um

7:37

go from kind of dreaming up a superpower to then refining the superpower and enabling the world to have it

7:44

very quickly like in a matter of like a few years and that's unprecedented now

7:50

when you can use that amazing software platform with like think of like ci cd and so on

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and what you're deploying is not just an app on your phone but you're deploying new coordination

8:02

mechanisms for organizing groups of people at various scales

8:08

you get the ability to like rewrite the planet right you can rewrite what we're all doing

8:14

and that's tremendously powerful if the the current blockchain networks are just

8:19

beginning to tap into the the potential here so this is why it's like super exciting

8:26

to be studying and working on crypto econ problems so i want to kind of like talk through a

8:32

very um specific component um that i've been um

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that a number of us have been like thinking about and so on and it like it's a very useful um

8:46

tool and we can use in the falcon community to potentially um

8:51

solve a bunch of problems so there's a class of mechanisms that we're calling impact evaluators

8:57

um and the way that we're sort of like this we still have to kind of better formalize these but

9:02

think of them as a simple mechanism that measures some impact and provide some reward

9:09

that's super general but that's kind of the point these can have a different frequency so these can be

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like a one-time process or it could be periodic this could have a scope that is

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proactive and or retroactive so proactive might mean that it um

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assesses potential impact and looks ahead and places some bets and rewards it ahead of

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time you can think of a grant system as an impact evaluator like this

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or it could be retroactive so you can look at some outputs measure those outputs measure the value of those outputs and reward proportionally

9:47

you can think of composability of these systems you can think of whether that reward schedule is

9:52

fixed or variable you can think of the incentive alignment between participants and

9:58

very specifically the the biggest crypto econ processes in the blockchain

10:05

space are extremely simple impact evaluators that are

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that part of why they work so well is that they are very stable and whole industries can be built

10:18

um relying on their structure so the bacon block reward is

10:24

the probably the the highest like um

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like highest impact impact evaluator uh in in the deployed so far um and it's

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it's kind of like absurd how much happens through this uh it's a very simple process all it does and it's kind of complicated how

10:42

it's implemented no um unfortunately the bitcoin uh community didn't kind of create a fully

10:48

programmable environment and so on so you don't have like a really nice um description of like here's an impact elevator and like here's what it's gonna

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do so you have to piece it together from what the protocol is doing but but at the end of the day if what's happening

11:00

is that you have a process that is measuring the hash rate contributed per miner per

11:06

block time and at every block time the ie is going to reward the miner with

11:12

probably with a probability proportional to the hash rate contributed and this is achieved by like you know doing the

11:17

whole um hash breaking and so on and so in expectation miners are getting

11:23

proportionally rewarded for their hashrate contributed and this impact evaluator alone has cost one of the

11:29

wildest energy-consuming processes on the planet um unfortunately most of

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that work is wasted um we're building falcon we want to do really useful things with that kind of process but this gives you a sense of

11:43

like how ies work and so you can then assess kind of like what are the properties of this thing

11:48

well it happens it has like a fixed schedule um it happens every 10 minutes it's running as part of the bitcoin

11:54

blockchain nobody can change it it's retroactive so it's not proactive

12:00

and its variability is that it's an exponentially decreasing emission but it's kind of discretized

12:06

and you can calculate it ahead of time so everybody knows what the entire reward schedule might be

12:12

and everybody also has a a view into all the hashrate contributed so far so people can make pretty

12:18

reliable predictions as to what the um future hatch rate contributions will be from

12:24

other parties and people can make pretty reliable predictions about their own contributions and so they can

12:31

make very reliable predictions about their own reward schedules when you couple that with the bitcoin price and

12:36

so on then you get out of that the entire bitcoin mining industry um

12:41

this in terms of incentive alignment this is zero sum so there's a fixed reward schedule and it um

12:47

it just decreases over time you could say that it's positive some because there's a secondary process here where

12:54

the this ie is actually causing like the purchasing of bitcoin so

13:01

it by feeding a lot of that economic activity into bitcoin it might it's kind

13:06

of like kind of positive sum um but that's a whole other whole other discussion

13:12

um you can think of the falcon block reward it's a very similar impact evaluator uh what is rewarding is qa

13:18

power contributor per block time here qa power is quality adjusted power that's where

13:24

we include capacity and um falcon plus verified storage

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and it rewards a set of miners with that same probability and so over time in expectation the falcon block rewards

13:37

rewarding miners proportionately um the incentive assignment actually there's the bug here it's

13:43

mixed because of the baseline

13:54

um you have again an exponentially decreasing emission in this case it's not discretized so it's every block or

14:00

against the to discretize whatever block is supposed to every four years and um

14:06

it's mixed because you have the baseline and the um above the baseline behavior so underneath the baseline it's um

14:13

it's positive sum above the baseline is zero sum and so on so what's really going on with these

14:19

things is that there's a very simple control theory loop so if you are familiar with uh robotics or anything like that like you

14:25

have a very straightforward structure for kind of there's some system

14:31

um that you can actuate and there's a the output of the system is fed into a sensor the sensor

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translates that into a feedback signal that goes into a kind of controller process and that gives some input into

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into the system so think of these kind of control theory loops as the kind of building blocks of impact

14:49

evaluators where you can think of the system as like the the network of participants the sensor as

14:56

the measured output that you see directly in the in the information on the chain or something like that and then the controller here

15:02

is the impact evaluator process that is deciding what the emission rate is or or whatever

15:08

um so this kind of like very simple uh structure could be used to reward all

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kinds of things in the network and this is one of these things that i think is vastly underutilized across our

15:22

our networks so for example and this is where i want to turn it into more of a discussion with everybody

15:28

i want you to think of different problems that we're dealing with in the falco network

15:34

and think of starting to construct impact evaluators for these because once we have programmability once that vm

15:40

lands we can and even today we can start with ethereum we can go and like create these in eth

15:45

um and start rewarding uh directly that way uh think of

15:51

some structure where all you want to do is have some process that you want to reward

15:57

and you want to create a periodic signal that you can draw out to then give a proportional reward to

16:03

participants for that process once you can frame it that way and declare it then you can kind of

16:08

start releasing currency and at the beginning you're not going to get much output but if it's reliable over time

16:14

that will translate into a lot of impact so maybe like think about some area of the

16:21

falco network this could be like we want hey maybe we want more crypto econ days maybe we want more crypto

16:27

conditions to happen all over the world we want people to learn about crypto econ maybe we want other kinds of events maybe you want

16:33

um we want to think about uh clients and client onboarding um

16:38

so we want to speed up the [Music] the onboarding rate of the network uh

16:45

maybe there could be an effective letter around that there sort of is one already but it's not as powerful as it perhaps could be

16:52

maybe we have what's the data onboarding

16:58

problem maybe we want to shuttle a bunch of drives from one place

17:04

to another in like physical machines how do we like could we turn that into an impact evaluator

17:10

um maybe we want more on ramp type things certain types of like material to show

17:16

up in the network could we create impact evaluators around that

17:22

and so to maybe like take some examples i want to kind of like work with this group to take an example and break it

17:28

down into something that we can create an impact evaluator over because if we get good at this then we can

17:33

steer the whole community towards like larger scale problems

17:39

so who has like a candidate problem that you want to solve in the network

17:44

raise your hand if you have a candidate problem you want to solve even if you don't know how to solve it yeah

17:51

i mean it's kind of one that you mentioned but onboarding and just to kind of put a finer point on it so we've

17:57

been me and steph mike dolinsky at the falkland foundation have been have been trying to work out some

18:03

ways of understanding this um and creating metrics for it essentially for usability right so one

18:11

of the things that's hard right now because it's not we're not really

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incentivized directly to tackle it as an ecosystem as a whole is simply you

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know downloading installing lotus setting it up and joining the network right it's it's it's hard and it's it's

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it's not it's no one's fault that it's hard right it's just that like there is no

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thing in the system that that's that specifically rewards that um

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one of the things that we can do and we do do and in some ways it's bad

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is we have multiple funnels to get to that point right if you search for how to install lotus or how to get vulcoin

18:54

running you'll see you'll hit five or six different ways of doing it um

19:00

so one thing that we can do is pull out like we have a measure we can measure

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between those right so we could pick out something in those systems that says

19:11

like one of them is like okay now going to this falcon faucet and pick up these

19:16

things and we're like oh okay we can now detect that someone is going through this funnel and we can also see whether

19:23

it trails off so anyway that's that's three percent of the way through the

19:28

kind of thinking that that we would need to do that um let's let's maybe um

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because that problem is very broad let's try and like decile it down how about like onboarding

19:39

um large clients so there's some set of clients out there right now we don't know we don't have like a a measure in

19:45

the chain of who are the potential clients we could get that so we could get some information to feed that into the chain

19:52

and we don't know what their user experience is right we don't have the way to rate the ux of onboarding the

19:59

data into the network the way that you get to rate a a lift car or an airbnb stay or

20:06

something like that so you could get some data some like qualitative data some quantitative and qualitative data

20:11

from clients and then integrate that into some feedback signal and feed it into some periodic reward

20:19

schedule um so what how would we like let's any

20:25

ideas on how we would do that how we like what do we need to add to the system to create a structure like this

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so we need some treasury with like some incentive to go towards some set of participants that are going to demonstrate that they've helped

20:38

onboard clients we have we need to get some data from

20:43

clients somehow and feed it into the chain as well some qualitative feedback output

20:50

so who can think through how to piece these things together for the people who report that the

20:56

onboard plan that they're telling you uh yeah so you do need to be able to

21:03

verify that the clients are telling art who they say they are and are telling

21:08

the truth yes yep so you need you need like the

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feedback signal that you're getting needs to be verifiable so that's certainly true

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but even like even before we get there how can we compose these signals to produce an

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impact evaluator the savings that these buyers have by

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switching you can measure the savings yeah um uh true so that would be like a

21:39

a competitive um analysis type of thing that one client that you could

21:45

advertise to the whole world once you're doing it well you can advertise that to the whole world and draw more clients

21:50

i'm thinking even more basic than this like you if i were to today like start programming a thing

21:55

what do i do what do i write like let's break it down into the component system so that i want to like get us all

22:01

thinking in like algorithmic terms so then you can start deploying these things

22:07

like how many new clients on board great so we need a set of how many clients are there out there potential

22:13

clients or actual clients and what their rate of change is right so we need to

22:19

know how much who the clients are we need to track them in the chain right now we don't know

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um they're not on there right like we know sort of but we know through mechanisms outside we don't know directly in the chain

22:30

so we need a set of clients and we need um to know how much storage they have an

22:36

interest in putting in or how much work they have already put in

22:42

what else we know like the um let's say we treat the data cap application right now as

22:48

like potential interest on the alarm yep and then you see how the throughput of how much electrical is being issued from an

22:54

apartment chain's perspective yep and then you also know how much has been consumed

23:00

so you have two flows the one with uh envelope that are kept exactly if you're consuming a data cap yep

23:06

so um so it's how you expect that to be a funnel so then we can establish let's say

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so we have this like inflow of data cap of like potential clients interest on the line right like you went through this process you get being captured on

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the chain maybe you're going to onboard 10 petabyte but at the same time you also know how fast this data card is

23:24

being consumed right like so and you say oh how many deals landed on the chain and what is the percentage of that but

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but right now um the data cap deployed to clients only sometimes is going to

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the actual clients sometimes it's going to the sps who are doing the work for the clients right so right now a lot of

23:40

sps are approaching clients and they're doing work with those clients

23:46

um and they're running the whole process for them like all the falcoin components so

23:52

um we we have to like get the actual intent from the client

23:58

and how much storage they might want to bring in and

24:04

um then if that's the case if i mean i feel like we're moving off the protocol then there are just a thousand there's many many ways to do that right just

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like no i'm saying like let's try and compose it into a protocol improvement like we could write a fip someday and

24:16

say let's add these set of components to improve the storage onboarding experience um oh sorry i have a

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hypothetical um it could be wrong like could be off but like maybe people can just like indicate that interest rates

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take some file coin and say hey i want to bring this bunch of a bunch of data onto the filecoin network that could now

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have some indicator on the on chain then you can say something so you have some indicator of interest and then you need

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to measure what do we like would you measure the output like it happening and some signal some qualitative or

24:47

quantitative signal out and feed it how do we do that what might

24:53

be a good way of like getting the feedback from the clients on how well that went

24:58

i have a different answer to a different question which this reminds me of the amazon the amazon affiliate program

25:04

right like you need some metadata there somewhere this is sort of trying to cure that the last thing which was like where

25:11

does the data cap live so right now we don't have anything on chain that says

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it came through this route right like so maybe like a tag that you would associate with data cap that was like i

25:24

it came through this funnel right so you have a destination yeah one of the larger meta points in a lot of what

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you're saying is that the client onboarding funnel is poorly instrumented in general

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and we need way better instrumentation of how that data is coming in to then over time

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measure what things are providing what outputs um and what quality what quality um what

25:50

kind of ex where are people getting stuck and why and so on so that's the point like maybe we should start by instrumenting the client funnel

25:57

figuring out all the different kinds of clients all the different groups and getting reliable and robust information

26:02

about their experience and how well they're progressing and once we have that then we can maybe

26:09

piece it into some impact evaluator we maybe started with one that was like

26:14

really hard important to us but really hard yeah

26:20

i mean so in this case the extra

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value that is created can actually be monetized in a sense right and it actually can be distributed

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um across you know all the people who contributed this is a easy monetization

26:38

problem um one of the i'm interested like

26:43

how would you do something like that if it's not as easily monetized so um

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yeah so um let's hold head in a totally different direction so i'm going to bring it back to impact evaluators

26:56

what i want us to be what i wanted this conversation to kind of get to is making things of this type because

27:02

these things are super easy to deploy and once they're deployed

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and they're stable and people can rely on them and they have data about their performance then people can start making

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predictions about the future and can start building entire businesses and industries over them so like you want to

27:20

get to something like as simple and straightforward as this deploy it um and

27:26

uh and enable a large community of participants to bet on that not changing so so like we gotta get to like simple

27:32

things like this maybe we can go from data onboarding to a different different problem

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what about the idea of um like an auto renewal like if somebody's having a good

27:42

experience they just oh my 12 month contract is over i'm just going to auto renew the exact same thing like that

27:48

would be kind of a vote it seems like a verifiable mechanical thing that's happening that

27:54

shows a vote of confidence or a thumbs up sort of thing yeah so um right now we don't have a

27:59

good way of like getting uh after the fact feedback from individual

28:05

participants and their experiences and that's kind of like uncommon in blockchains but very common

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in our day-to-day activity on tons of applications right like car services hotels all of these

28:19

things um browsers like everything asks you for your feedback and like they start rating

28:24

your output and based on that they use that signal to optimize their system somewhere so maybe we should just that

28:31

might be like a very trivial ad like find a good way to contribute um to be able to pull

28:38

participants for feedback we have to make it verifiable in some way to the point before like we have to make sure that that data's of high quality but

28:45

that might give us a lot of signal so how about let's let's think about events suppose that we want to have

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we're like planning next year's events and we want to have tons of community organized events around the world how might we design an

28:57

impact evaluator to cause that

29:04

yep i guess you have to work out what you want from the events

29:10

and often it's about trying to bring people into community hire people so

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if that's if you measure how many people you're actually hiring based on the attendance of the event you can start to

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yeah measure the impact yeah so so um that's like a like a specific

29:27

um thing around you have a certain set of qualities that you want to gauge about the event uh you have one

29:34

you describe like one measurable output from events like people hired um you can also measure attendance you can also

29:41

measure like locations you can also measure like um

29:50

yeah do you want to tell us

30:07

you could talk about media mentions yeah um you could talk about recognizability of

30:15

file coin or whatever the thing is that the event is so we have a bunch of signals that a lot of different groups care

30:20

about of what is a good event and we need to be able to get some rel reliable enough measurement it doesn't

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have to be perfect but it has to be reliable enough to then feed back into a signal so suppose that we can do that suppose that

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we have these like set of measures and we can after an event happens go back and score them for each one of

30:39

these i'm convinced that all of us can come together and come up with pretty good ways of getting that data out of

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the internet and come up with a score suppose that we now have a set of events that are happening

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with like some set of you know properties and their scores how we how do we go from that into an

30:56

impact evaluator

31:02

we look at the frequency that we're going to measure it by yeah exactly so what frequency let's propose them

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probably directly before the event directly after the event and you know like a month

31:16

after yeah so we so we could we could design an impact evaluator that as every event

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happens measures that event so you could have like a proactive grant making thing that gives some

31:27

capital for people to try making putting on an event you can measure the outputs and then reward it retroactively

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how about we do something like what the what these block rewards do which is kind of like um

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what's extremely powerful about these is that they're auctions so the way that they work is that they

31:44

basically walk the impact evaluator walks up to an open network and says

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um there's this amount of reward i will deploy this amount of reward in this unit time

31:55

to proportionally to the overall total quality that i get

32:00

out and that that is able to like enable all participants to put in

32:06

whatever they think um they want to add for whatever the reward is worth um i'm

32:12

not saying that this well but like auctions tend to be like a really good way to kind of let um participants in a

32:18

network find like the best price yep what comes to mind is like

32:24

a lottery or something so it's like you want to maybe keep incentivize people that add all this feedback i enjoyed this event i like this session i didn't

32:31

like that session and the more feedback you give the more like lottery tickets you get for some sort of

32:38

reward at the end so like might incentivize people to show up in the first place might incentivize people to get feedback along the way

32:44

and it everyone has it sort of reminds me of the bitcoin blocker award it's sort of like a someone gets all the value but

32:50

everyone's motivated to yeah so that feedback would be all probably feed into one of

32:56

the qualities that we're measuring about an event right so let's write down

33:12

so i think we're talking about this in terms of like a very like automated system

33:18

um and so like there's you know there's a curious on your thoughts of weighing the

33:23

value of like a rapid prototype um doing something very like manually because you

33:29

don't know what inputs you're actually going to care about you want to just quickly go through a bunch of them i'm

33:36

thinking more about data onboarding right now verse having something that is more

33:43

programmed and automated that

33:49

makes sure that you are not like like when humans are doing it

33:55

just like thinking about things there's always the temptation to not be very stringent

34:01

about what things you're evaluating um where if you're doing it more

34:06

programmatically it's like more well-defined what are your thoughts on that one um

34:12

so i think that what's extremely useful about programmatic structures that

34:18

are regular and dependable in the long term is that you decrease a transaction cost

34:25

of entering into those conversations however they're much less friendly and forgiving right

34:32

for example in a grant program there's all kinds of back and forth that are instrumental in refining grants into a

34:38

good structure that is likely to produce really good value so um it's not clear yet that you can have a good automated

34:45

grant making thing that isn't like totally like immediately produces like bad output and is like incentive misaligned and so on so so i think that

34:52

there are a lot of things that that do require um a lot of individual um

34:57

involvement in kind of like human oriented programs um but the transaction cost is very high

35:03

for each one of those things you're going to have a very high transaction cost of being able to interact being able to

35:10

describe the the potential value and align together onto a good outcome and what's extremely powerful about

35:17

these like ies in the cloud so to speak is that once you put them out there

35:22

participants can transact directly with the ie and you drop you drop the transaction costs between all

35:28

participants and so you create a much more open environment and so

35:33

they don't work well for things that are very fuzzy and hard to quantify

35:39

so things that are really qualitative or where you are trying to align on like potential predictive value that they

35:44

don't tend to work really well but for very concrete things that you can turn into measurable this like concrete quantified

35:52

impact over time they can work really well for that so and i do think that we can even with something like events that are

35:58

ex so so extremely qualitative um you can get some measurable outputs out of that and turn that into enough of

36:05

a signal to to know what events in that in the region of in a period of time were kind of like regarded by the

36:11

community as like the most valuable so to speak um and then out of that you can like then create kind of an impact about

36:17

later type things relatedly what are your thoughts is not to just take this over but uh what are your thoughts on like

36:22

overfitting then to data that is already measurable rather than creating new mechanisms of measuring because you're

36:29

sort of like always tempted then to just like use the 100 of these over fit so these these you you get exactly what you

36:35

reward so like totally you like uh you know um so the bitcoin community did not want to

36:42

maximize the amount of hash rate on the world right like um can't really speak for associate here

36:48

but i really don't think satoshi and the bitcoin community sat around thinking you know how can we

36:54

turn the entire planet into hash rate um and like put every single computer into an asic

37:00

um and so 100 these are fit and that's a misapplication of an ie right

37:06

like releasing this thing into the world created like this

37:11

massive consumption of energy in like this like wasteful process and now it's like we have to

37:17

fix that and we have like this runaway process that we have to like go and fix so we've learned since then like

37:24

we're making better things that are like now more attached to value um so for example the the

37:30

falcon plus useful storage thing is like a huge upgrade on on just like say just capacity

37:36

or even that is a capacity um is a huge upgrade on like wasted hash rate so can

37:43

we kind of like go to some things that like there are some domains where this kind of thing can be very helpful there are

37:50

many cases where they won't be and like you you don't want to kind of incentivize something to overfit too much

37:56

but there are many problems with at enough scale that you can turn it into like pretty good quantifiable

38:02

output that you can then reward this way it's kind of like what i think it's like for some problems you can use these for

38:08

some from a lot of problems you can yeah

38:14

can we quickly just go back to danny's question and i'd love to hear your thoughts uh can this impact you know onboarding

38:21

clients right can we use like unique data set maybe we use some algorithm that matches and say if it's 80

38:27

different it's a unique data set and then the variability will be like increasing uh instead of decreasing

38:33

because as you onboard more and more then it's harder to find unique data set would that be something that solves

38:39

partially yes so um yeah as possible i mean i think like you could say hey there's a class of

38:45

data that we want to onboard suppose that we say for the atlas project that you'll hear about um

38:50

you want um you want to get like geospatial data onto file coin and like right now you

38:55

want to run an ie to get like pictures of the planet like added at different resolutions

39:01

um so what you can do is like you can set up an impact elevator to reward participants that contribute

39:08

and that put into the network like pictures of the world taken between you know one time and another time you end

39:14

up in a hard problem where like you have to verify those things like you have to verify that that was corr done correctly but

39:20

you can at that point reward all participants for bringing that specific type of data into the network and it will cause a lot

39:27

of it to appear but you better be good about like your verifiability because you're gonna get the wrong output like you'll over fit like the network will

39:33

overfit to what your reward says not to the intent of your reward right like this is where like the letter of the law

39:39

really matters if you get that wrong like you'll you'll get the wrong output

39:45

thank you yeah no no we're good no no i'm happy to like pause i wanted to get people

39:50

thinking about impact evaluators i think like um they're not they're uh

39:56

non-trivial systems what i kind of want to leave you with is like um you can generally take these larger

40:02

scale problems breaking them into smaller ones and then if you can come up with a concrete way to measure

40:08

the output and feed it into a periodic reward schedule um

40:14

then you can like then you can move mountains but you better

40:19

to the point you better like know what you're measuring all right thanks

40:43

you